Isaac G - Derivation of the Magnitude Uncertainty from Signal-to-Noise Ratio

The astronomical magnitude of a source with measured flux F is defined via Pogson's relation [1]:

$$m = -2.5 \log_{10}(F) \tag{1}$$

For a value m that is a function of a measured quantity F with uncertainty σ_F, the propagated uncertainty σ_m is, to first order (standard linear error propagation, as shown in the sources [2, 3]):

$$\sigma_m \approx \left| \frac{dm}{dF} \right| \sigma_F \tag{2}$$

I got this from a first-order Taylor expansion of $m(F)$ about the measured value of F (had to just look this up in wolfram alpha), and is valid provided $\sigma_F \ll F$.

Rewrite Eq. (1) using the change-of-base identity $\log_{10}(F) = \ln(F)/\ln(10)$:

$$m = -\frac{2.5}{\ln(10)} \ln(F) \tag{3}$$

Differentiating with respect to F, using $\frac{d}{dF}\ln(F) = \frac{1}{F}$:

$$\frac{dm}{dF} = -\frac{2.5}{\ln(10)} \cdot \frac{1}{F} \tag{4}$$

Substituting Eq. (4) into Eq. (2):

$$\sigma_m = \frac{2.5}{\ln(10)} \cdot \frac{\sigma_F}{F} \tag{5}$$

The quantity σ_F/F is the fractional (relative) flux uncertainty, which is the reciprocal of the signal-to-noise ratio, $\mathrm{SNR} \equiv F/\sigma_F$ [4, 5]:

$$\frac{\sigma_F}{F} = \frac{1}{\mathrm{SNR}} \tag{6}$$

Substituting into Eq. (5):

$$\boxed{\sigma_m = \frac{2.5}{\ln(10)} \cdot \frac{1}{\mathrm{SNR}} \approx \frac{1.0857}{\mathrm{SNR}}} \tag{7}$$

since

$$\frac{2.5}{\ln(10)} = \frac{2.5}{2.302585\ldots} = 1.085736\ldots \tag{8}$$

Note on validity

Equation (7) is a first-order (linearized) approximation and is most accurate when $\sigma_F \ll F$, i.e. at high SNR. Near a detection threshold (e.g. SNR ~ 5), the fractional flux error is no longer small, and the linear approximation could be biased. I did not notice this bias so far, but I will continue to check it.

References

[1] Pogson, N. R. (1856). Magnitudes of Thirty-six of the Minor Planets for the first day of each month of the year 1857. *Monthly Notices of the Royal Astronomical Society*, 17, 12–15.

[2] Taylor, J. R. (1997). *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements* (2nd ed.). University Science Books.

[3] Bevington, P. R., & Robinson, D. K. (2003). *Data Reduction and Error Analysis for the Physical Sciences* (3rd ed.). McGraw-Hill.

[4] Howell, S. B. (2006). *Handbook of CCD Astronomy* (2nd ed.). Cambridge University Press.

[5] Mighell, K. J. (1999). Algorithms for CCD Stellar Photometry. In *Astronomical Data Analysis Software and Systems VIII*, ASP Conference Series, 172, 317.

[6] Merline, W. J., & Howell, S. B. (1995). A Realistic Model for Point-sources Imaged on Array Detectors: The Model and Initial Results. *Experimental Astronomy*, 6, 163–210.

Note: These are the sources taken from the Foundations of Astrophysics textbook by Barbara Ryden. That's the textbook that I based all of this on.